

# Aayaan Sahu

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## EDUCATION

### University of Illinois Urbana-Champaign

*B.S. Computer Science + Economics*

**GPA:** 4.0

**Clubs:** Hack4Impact, Disruption Lab, Illinois Space Society

Champaign, IL

*Expected Graduation: May 2028*

## SKILLS

**Languages:** Python, TypeScript, C++, R, SQL, Java, Go

**Libraries/Tools:** PyTorch, LangGraph, Scikit-learn, Pandas, Numpy, OpenCV

**Frameworks:** JAX, FastAPI, React, Django, Flask, Next.js, Express

**Systems:** Transformers, Diffusion, RAG, Vector Databases, AWS, Git, Docker

## EXPERIENCES

### Researcher — Concept Editing in Vision Models

Feb 2026 - Present

*Advised by Dr. Ahuja of Computer Vision Robotics Lab*

*Champaign, IL*

- Trained ResNet and ViT via data normalization and augmentation with **96% accuracy** on 100+ class datasets
- Isolated Concept Activation Vectors through supervised methods (TCAV / ACE) and unsupervised methods (ICE)
- Extracted human-interpretable representations from deep networks by optimizing gradients, leveraging **FFT output in complex plane** to solve noisy gradient problem
- Developing novel architecture consisting of paired ResNet and GAN to enable "concept editing" via targeted modification of latent embedding

### Researcher — Analyzing Bias in Emergency Room Triage

Jun 2021 - Dec 2022

*Aspiring Scholars Directed Research Program*

*Fremont, CA*

- Preprocessed 10+ large scale US medical system ER datasets (> 1M rows) to predict triage outcomes
- Trained and evaluated Random Forest, XGBoost, and neural networks to predict triage outcomes
- Implemented SHAP values for interpretability, identifying statistically significant model bias factors
- Published in **Annals of Biomedical Science and Engineering** and **Stanford's JUST Health** (pg. 13)
- Presented findings at Southern California Conference for Undergraduate Research (11/19/22)

## PROJECTS

### PRISM | *AI Powered Personalized Real-time Identified Speech Masking*

Feb 2026 – Present

- Architected a **real-time voice intelligence agent**, using novel sliding window-technique, leveraging Websockets and multithreading for concurrent target-speaker extraction, isolating select audio sources for noisy environment
- Implemented **Lip Decoder audio-visual ML model** for filtering target voice, pairing with speech embedding, and target speaker extraction models
- Achieved **industry standard ~300 ms** end-to-end latency with cloud GPU inference, matching production-grade voice systems

### LLMBDA | *Prompt Optimization Tool*

Nov 2025 – Present

- Optimized LLM prompt efficiency by achieving **20x token compression** via custom task router and research-backed text compression techniques
- Engineered and launched a production-ready SaaS platform with React and Stripe, managing the end-to-end lifecycle from development to a successful Chrome Web Store launch with **30+ users**

### Kova | *AI-Powered Real-Time Scam Call Protection*

Jan 2026 – Feb 2026

- Developed an autonomous scam-detection agent by implementing a LangGraph state machine for real-time risk assessment, utilizing low-latency transcription
- Designed a low-latency two-stage transcription pipeline (Deepgram STT → LLaMA 3.3 70B speaker diarization) powered with FastAPI and Websockets
- Implemented a multi-state alert system based on risk and confidence thresholds, achieving **90% accuracy** on scam detection

### TalkPDF | *Retrieval-Augmented Generation System*

Jun 2025 - Aug 2025

- Engineered a full-stack web app integrating Python, **SQL**, Supabase, Stripe, **Ollama**, and **vector databases**
- Implemented document ingestion, cleaning, and querying to allow users to "chat with a PDF"
- Designed clean **React** frontend with ShadCDN and ensured seamless UX across browsers